# Capture the Flag (CTF) Report

**Internship Assignment – HackTheBox Starting Point Machines**
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## Introduction

This report documents the successful completion of a Capture the Flag (CTF) assignment during my cybersecurity internship. The objective was to capture at least three flags in HackTheBox (HTB) and provide a clear writeup of the methodology.
I successfully captured flags from three machines: Meow, Fawn, and Dancing. Each machine demonstrated a common misconfiguration or vulnerability (Telnet with no password, FTP with anonymous login, SMB with open shares).
The findings highlight how insecure defaults can expose sensitive data. This report explains the process step-by-step, with screenshots, so that even non-technical readers can understand the risks and lessons learned.

## Environment Setup

**Operating System**: Kali Linux (VMware Workstation).
**Connection:**Pwnbox and  HackTheBox VPN using OpenVPN

## Tools Used:

**Nmap** – scans systems for open ports.
**Ftp** – connects to FTP servers.
**Smbclient** – connects to Windows file shares.
Standard Linux commands; **ls, cat and get**
Screenshot of my Kali VM connected to HTB VPN below

```
kali@kali: ~/D
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/Downloads]
└─$ sudo openvpn htb.ovpn
2026-03-26 12:11:39 DEPRECATED OPTION: --persist-key option ignored. Keys are now
2026-03-26 12:11:39 WARNING: Compression for receiving enabled. Compression has be
commend to disable it completely.
2026-03-26 12:11:39 Note: --data-ciphers-fallback with cipher 'AES-128-CBC' disabl
2026-03-26 12:11:39 OpenVPN 2.7.0 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4]
2026-03-26 12:11:39 library versions: OpenSSL 3.5.5 27 Jan 2026, LZO 2.10
2026-03-26 12:11:39 DCO version: N/A
2026-03-26 12:11:41 TCP/UDP: Preserving recently used remote address: [AF_INET]38.
2026-03-26 12:11:41 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-03-26 12:11:41 UDPv4 link local: (not bound)
2026-03-26 12:11:41 UDPv4 link remote: [AF_INET]38.46.224.104:1337
2026-03-26 12:11:41 TLS: Initial packet from [AF_INET]38.46.224.104:1337, sid=7827
2026-03-26 12:11:42 VERIFY OK: depth=2, C=GR, O=Hack The Box, OU=Systems, CN=HTB V
2026-03-26 12:11:42 VERIFY OK: depth=1, C=GR, O=Hack The Box, OU=Systems, CN=HTB V
2026-03-26 12:11:42 VERIFY KU OK
2026-03-26 12:11:42 Validating certificate extended key usage
2026-03-26 12:11:42 ++ Certificate has EKU (str) TLS Web Client Authentication, ex
2026-03-26 12:11:42 ++ Certificate has EKU (oid) 1.3.6.1.5.5.7.3.2, expects TLS We
2026-03-26 12:11:42 ++ Certificate has EKU (str) TLS Web Server Authentication, ex
2026-03-26 12:11:42 VERIFY EKU OK
2026-03-26 12:11:42 VERIFY OK: depth=0, C=GR, O=Hack The Box, OU=Systems, CN=us-st
2026-03-26 12:11:42 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA38
 ed25519 ED25519, key agreement: X25519MLKEM768
2026-03-26 12:11:42 [us-starting-point-1-dhcp] Peer Connection Initiated with [AF_
2026-03-26 12:11:42 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2026-03-26 12:11:42 TLS: tls_multi_process: initial untrusted session promoted to
2026-03-26 12:11:43 SENT CONTROL [us-starting-point-1-dhcp]: 'PUSH_REQUEST' (statu
2026-03-26 12:11:44 PUSH: Received control message: 'PUSH_REPLY,route 10.10.8.0 25
38.0 255.255.255.0,route 10.10.110.0 255.255.255.0,route-ipv6 dead:beef::/64,expli
rt 120,ifconfig-ipv6 dead:beef:2::1022/64 dead:beef:2::1,ifconfig 10.10.14.36 255.
,tun-mtu 1500'
2026-03-26 12:11:44 OPTIONS IMPORT: --ifconfig/up options modified
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.
```

# Methodology

The same structured approach was applied to each machine:

**Reconnaissance**: Scan the target with  nmap to identify open ports.

**Enumeration**: Investigate services running on those ports.

**Exploitation:** Use weak/default credentials or misconfigurations to gain access.

**Flag Retrieval:** Locate and download the hidden flag.txt  file.

**Documentation**: Record commands, outputs, and screenshots.

Screenshot of an nmap scan result here.

```
└─[★]$ nmap -sV 10.129.55.103
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-26 06:29 CDT
Nmap scan report for 10.129.55.103
Host is up (0.089s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
23/tcp open  telnet  Linux telnetd
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.13 seconds
┌─[us-starting-point-1-dhcp]─[10.10.14.36]─[tony000123@htb-kbynqnsou3]─[~]
└─[★]$
```

3 items

Menu [VNC config] Computer Parrot Terminal

https://vnc.htb-cloud.com/index.php

Applications Places System Thu Mar 26, 05:11

Parrot Terminal

File File Edit View Search Terminal Help

```
┌─[us-starting-point-1-dhcp]─[10.10.14.36]─[tony000123@htb-kbynqnsou3]─[~]
└─[★]$ nmap -sV 10.129.55.57
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-26 05:10 CDT
Nmap scan report for 10.129.55.57
Host is up (0.090s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.3
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.14 seconds
┌─[us-starting-point-1-dhcp]─[10.10.14.36]─[tony000123@htb-kbynqnsou3]─[~]
└─[★]$
```

3 items

Menu [VNC config] Computer Parrot Terminal

```
┌──(kali㉿kali)-[~/Downloads]
└─$ nmap -sV 10.129.64.75
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-27 06:52 -0400
Nmap scan report for 10.129.64.75
Host is up (0.92s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
135/tcp  open  msrpc         Microsoft Windows RPC
139/tcp  open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp  open  microsoft-ds?
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 48.04 seconds

┌──(kali㉿kali)-[~/Downloads]
└─$
```

# My Findings Per Machine

## Meow

**Service**: Telnet (port 23).
**Weakness**: No password required.
**Process**: Connected via Telnet, gained shell access immediately.
**Flag Retrieval**: Navigated to the home directory and found flag.txt
**Lesson Learned**: Default/no-password services are a major risk.
screenshot of Telnet connection and flag retrieval here.

https://vnc.htb-cloud.com/index.php

Applications Places System Thu Mar 26, 06:39

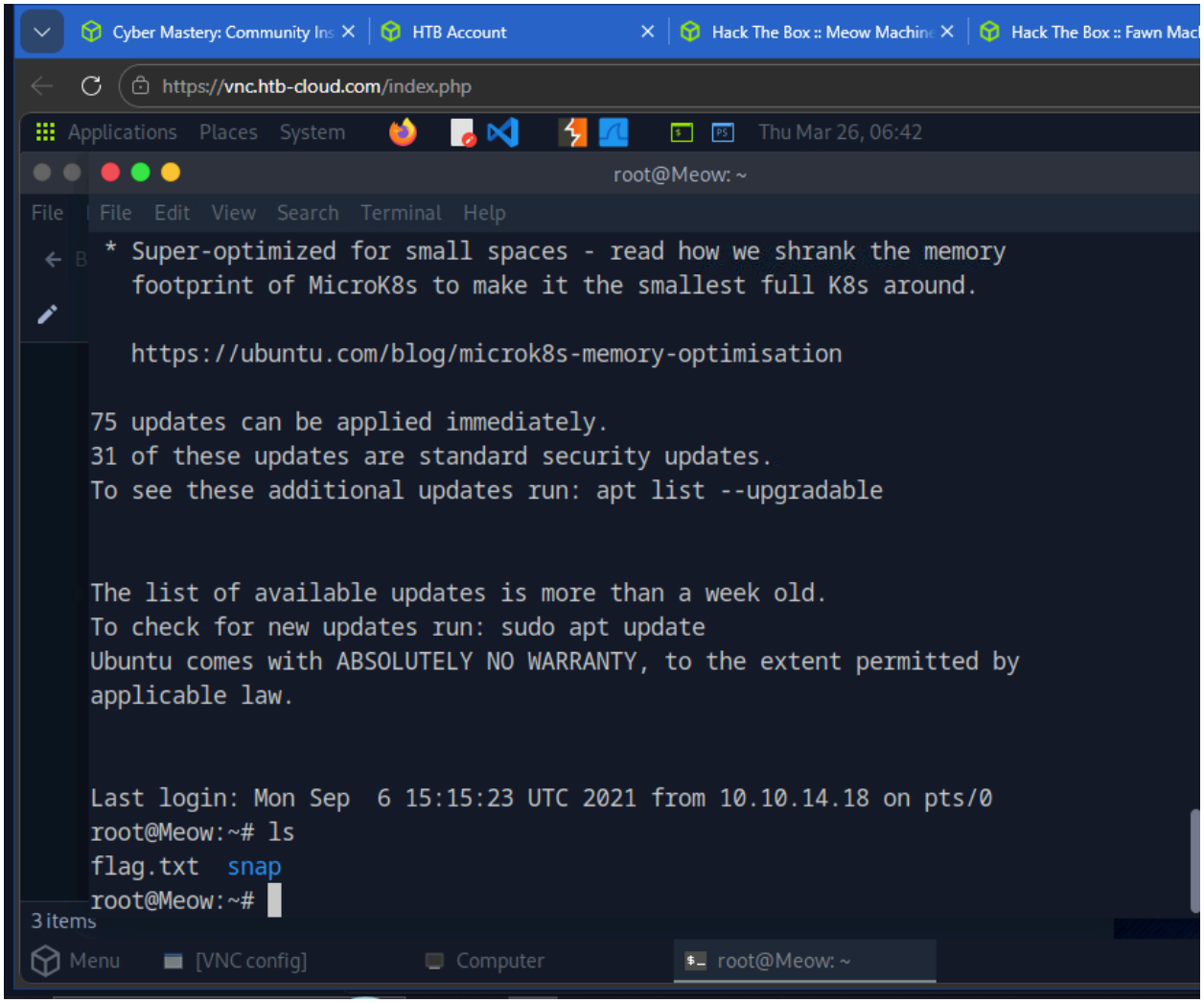
root@Meow: ~

File Edit View Search Terminal Help

```
└─[us-starting-point-1-dhcp]─[10.10.14.36]─[tony000123@htb-kbynqnsou3]─[~]
└──╼ [★]$ root
bash: root: command not found
┌─[us-starting-point-1-dhcp]─[10.10.14.36]─[tony000123@htb-kbynqnsou3]─[~]
└──╼ [★]$ telnet 10.129.55.103
Trying 10.129.55.103...
Connected to 10.129.55.103.
Escape character is '^]'.

Hack the Box

Meow login: root
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-77-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
```

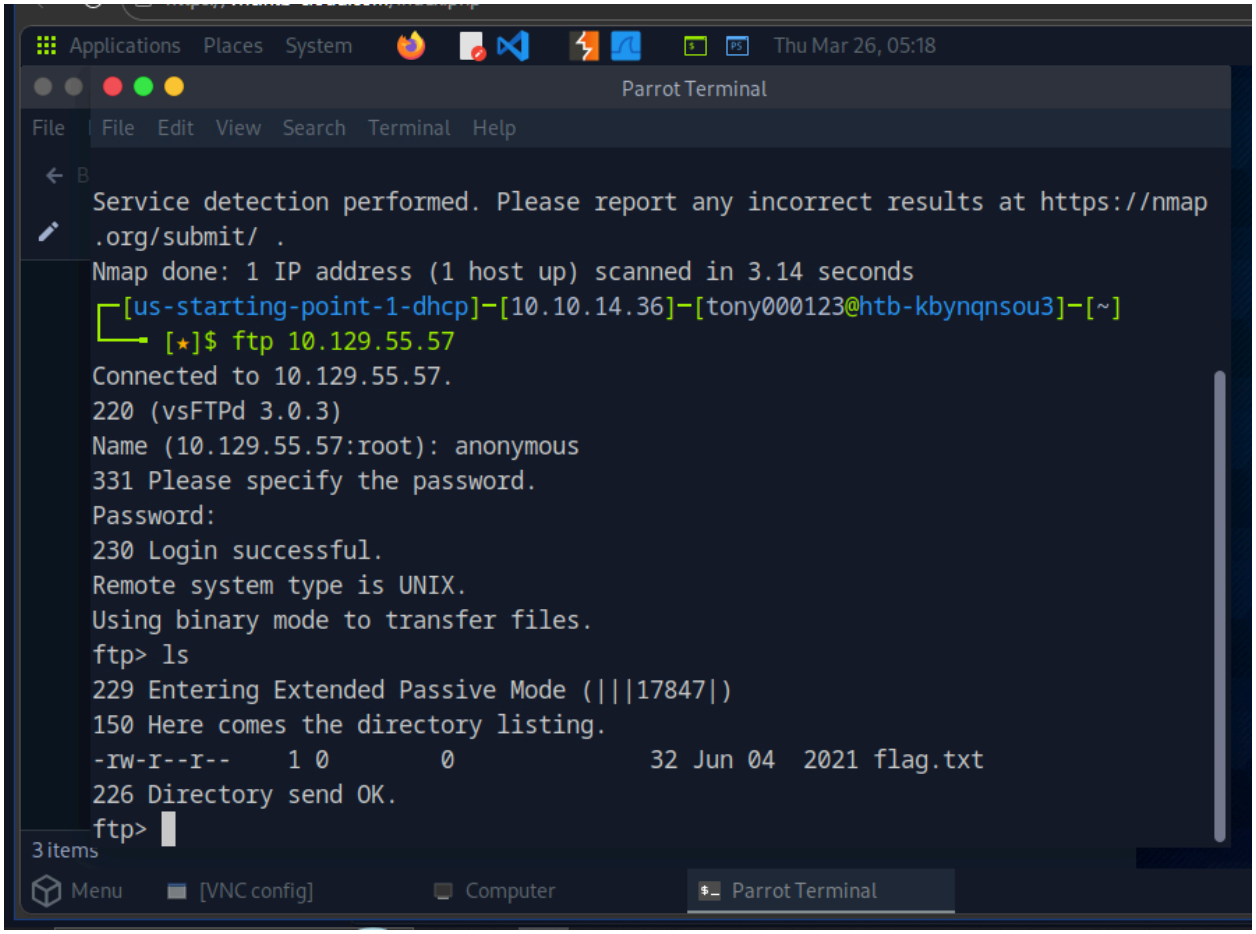
3 items

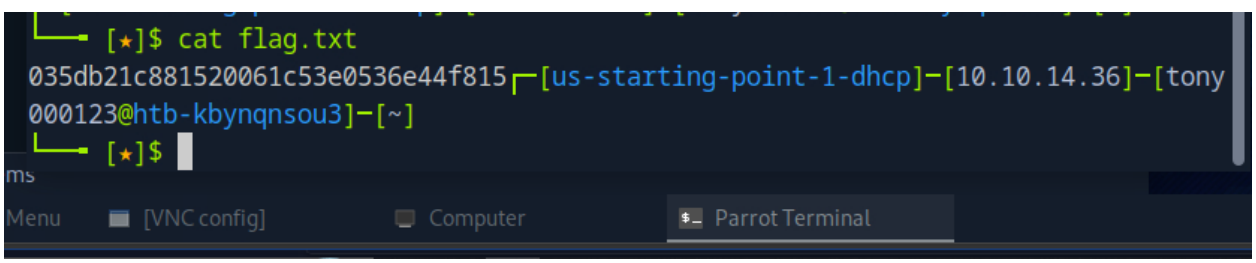
Menu [VNC config] Computer root@Meow: ~




## Fawn

**Service:** FTP (port 21).
**Weakness**: Allowed anonymous login.
**Process:** Logged in with username anonymous and no password.
**Flag Retrieval:** Listed files, downloaded flag.txt
**Lesson Learned:** Anonymous FTP access exposes sensitive files.
screenshot of FTP login and flag download here.

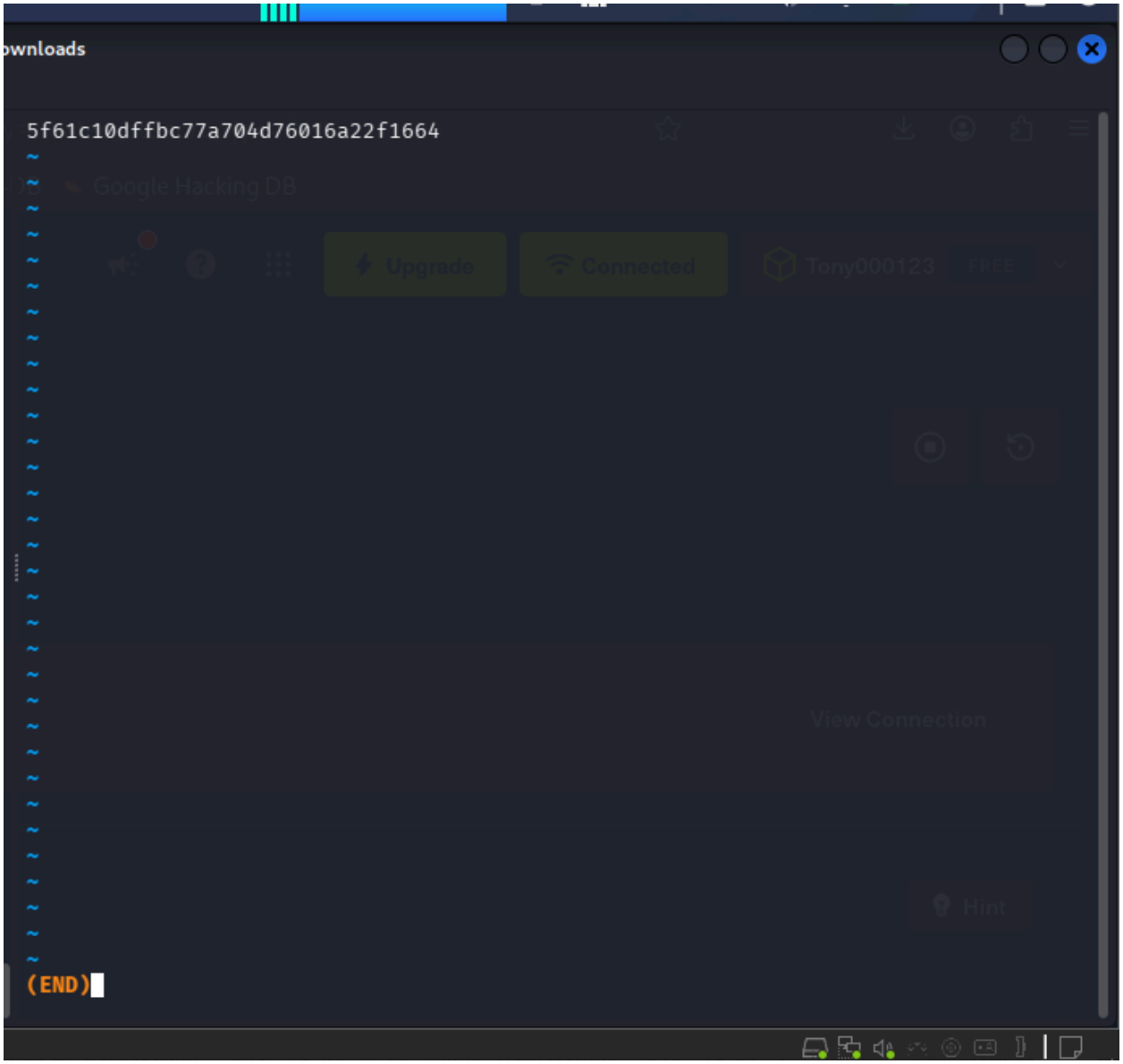



## Dancing

**Service**: SMB (port 445, Microsoft-DS).
**Weakness**: Shared folder accessible with blank password.
**Process**: Connected with smbclient//IP/Workshares -N, navigated into james.P **Process:** Connected with `smbclient //IP/WorkShares -N`, navigated into `James.P`, and Downloaded flag.txt
**Flag Retrieval:** Used, get flag.txt  inside the SMB shell.
**Lesson Learned:** Misconfigured file shares can leak confidential data.
*Screenshot of SMB connection, directory listing, and flag download here.*

```
Service detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 48.04 seconds

┌──(kali㉿kali)-[~/Downloads]
└─$ smbclient -L \\10.129.64.75 -N

        Sharename       Type      Comment
        ---------       ----      -------
        ADMIN$          Disk      Remote Admin
        C$              Disk      Default share
        IPC$            IPC       Remote IPC
        WorkShares      Disk
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.64.75 failed (Error NT_STATUS_RESOURCE_NAME_NOT_F
OUND)
Unable to connect with SMB1 -- no workgroup available

┌──(kali㉿kali)-[~/Downloads]
└─$ smbclient \\10.129.64.75\WorkShares -N

\10.129.64.75WorkShares: Not enough '\' characters in service
Usage: smbclient [-?EgqBNPkV] [-?|--help] [--usage] [-M|--message=HOST]
        [-I|--ip-address=IP] [-E|--stderr] [-L|--list=HOST]
        [-T|--tar=<c|x>IXFvgbNan] [-D|--directory=DIR] [-c|--command=STRING]
        [-b|--send-buffer=BYTES] [-t|--timeout=SECONDS] [-p|--port=PORT]
        [-g|--grepable] [-q|--quiet] [-B|--browse]
        [-d|--debuglevel=DEBUGLEVEL] [--debug-stdout]
        [-s|--configfile=CONFIGFILE] [--option=name=value]
        [-l|--log-basename=LOGFILEBASE] [--leak-report] [--leak-report-full]
        [-R|--name-resolve=NAME-RESOLVE-ORDER]
        [-O|--socket-options=SOCKETOPTIONS] [-m|--max-protocol=MAXPROTOCOL]
        [-n|--netbiosname=NETBIOSNAME] [--netbios-scope=SCOPE]
        [-W|--workgroup=WORKGROUP] [--realm=REALM]
```




## *Conclusion*

I Captured three flags (Meow, Fawn, Dancing) Demonstrated practical penetration testing methodology: reconnaissance, enumeration, exploitation, and flag capture. Learned how insecure defaults (no password, anonymous access, open shares) can compromise systems. This exercise strengthened my ability to document findings clearly and communicate technical work in a way that non-technical stakeholders can understand. In addition, my recommendation for any organisation includes, always disable insecure services (Telnet, anonymous FTP). Always require strong authentication for all network services.
Regularly audit file shares and access permissions and .Use vulnerability scanning tools to identify misconfigurations early

# Deliverables

Flags: Meow, Fawn, Dancing.
Writeup document (this report).
Screenshots of methodology and flag submissions.